

# MATH 39 — Linear Algebra

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# 1. Chapter 1: Systems of Linear Equations and Matrices

## 1.1 Linear Equations

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**Definition** A **linear equation** in the variables  $x_1, x_2, \dots, x_n$  is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where  $a_1, a_2, \dots, a_n$  and  $b$  are constants. The constants  $a_1, a_2, \dots, a_n$  are called the **coefficients** of the equation, and  $b$  is called the **constant term**.

**Examples** of linear equations include:

- $2x + 3y - z = 5$
- $-x_1 + 4x_2 + 0x_3 = 7$
- $0x + 0y + 0z = 0$

**Non-examples** of linear equations include:

- $x^2 + y = 3$  (because of the  $x^2$  term)
- $xy + z = 4$  (because of the  $xy$  term)
- $\sin(x) + y = 1$  (because of the  $\sin(x)$  term)
- $x + \sqrt{y} = 2$  (because of the  $\sqrt{y}$  term)

## Systems of Linear Equations

**Definition** A **finite** collection of linear equations involving the same set of variables is called a **system of linear equations**.

**Example** of a system of linear equations:

$$\begin{cases} 2x + 3y - z = 5 \\ -x + 4y + 2z = 7 \\ 3x - y + z = 2 \\ \dots \end{cases}$$

**Definition** A sequence of numbers  $(s_1, s_2, \dots, s_n)$  is called a **solution** to the system of linear equations if, when we substitute  $s_i$  for  $x_i$  in each equation, the equation holds true.

**Example** Consider the system of equations:

$$\begin{cases} x + y + z = 3 \\ 2x + y + 3z = 1 \end{cases}$$

Is the sequence  $(-2, -5, 0)$  a solution to this system? Does  $x = y = z = 1$ ?

**Solution** To check if  $(-2, -5, 0)$  is a solution, substitute  $x = -2$ ,  $y = -5$ , and  $z = 0$  into both equations:

- First equation:  $(-2) + (-5) + 0 = -7 \neq 3$  (does not hold)
- Second equation:  $2(-2) + (-5) + 3(0) = -4 - 5 = -9 \neq 1$  (does not hold)

Thus,  $(-2, -5, 0)$  is not a solution to the system.

Next, check if  $(1, 1, 1)$  is a solution:

- First equation:  $1 + 1 + 1 = 3$  (holds)
- Second equation:  $2(1) + 1 + 3(1) = 2 + 1 + 3 = 6 \neq 1$  (does not hold)

Thus,  $(1, 1, 1)$  is also not a solution to the system.

## Solutions to Linear Systems

**Definition** There are three possible scenarios for the solutions to a system of linear equations:

- **No solution:** The system is inconsistent, meaning there are no values for the variables that satisfy all equations simultaneously.
- **Unique solution:** There is exactly one set of values for the variables that satisfies all equations.
- **Infinitely many solutions:** There are multiple sets of values for the variables that satisfy all equations. (i.e. the same line or scalar multiples of each other)

Systems of linear equations that have at least one solution are called **consistent**, while those with no solutions are called **inconsistent**.

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## Matrices

**Definition** A **matrix** is a rectangular array of numbers arranged in rows and columns. The numbers in the matrix are called its **entries** or **elements**. A matrix with  $m$  rows and  $n$  columns is called an  $m \times n$  matrix.

**Example** of an **augmented** matrix:

$$\begin{cases} 3x_1 + 2x_2 - x_3 + x_4 = -1 \\ 2x_1 - x_3 + 2x_4 = 0 \\ 3x_1 + x_2 + 2x_3 + 5x_4 = 2 \end{cases} = \left[ \begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array} \right]$$

## Elementary Operations

**Definition** There are three types of **elementary row operations** that can be performed on a matrix:

- **Row swapping:** Interchanging two rows of the matrix.
- **Row scaling:** Multiplying all entries in a row by a non-zero constant.
- **Row addition:** Adding a multiple of one row to another row.

These operations are used in methods such as Gaussian elimination to solve systems of linear equations. Row operations do not produce equal matrices, but they do produce **equivalent** matrices, meaning they represent systems of equations with the same solution set. In order for 2 matrices to be equal, they must have the same dimensions and identical entries in corresponding positions.

**Example** Consider the matrices:

$$\left[ \begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 4 & 1 \end{array} \right] \mapsto \left[ \begin{array}{cc|c} 2 & 4 & 0 \\ 3 & 4 & 1 \end{array} \right]$$

This represents the row operation of scaling the first row by 2 ( $2R_1 \mapsto R_1$ ). These matrices are equivalent but not equal.

**Example** Solve:

$$\begin{aligned} \begin{cases} 2x + 3y - z = 3 \\ -x - y + z = 1 \\ x - 2y - 5z = 0 \end{cases} &= \left[ \begin{array}{ccc|c} 2 & 3 & -1 & 3 \\ -1 & -1 & 1 & 1 \\ 1 & -2 & -5 & 0 \end{array} \right] \xrightarrow{R3 \mapsto R1} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ -1 & -1 & 1 & 1 \\ 2 & 3 & -1 & 3 \end{array} \right] \xrightarrow{R2+R1 \mapsto R2} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 2 & 3 & -1 & 3 \end{array} \right] \\ &\xrightarrow{R3-2R1 \mapsto R3} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & -3 & -4 & 1 \\ 0 & 7 & 9 & 3 \end{array} \right] \xrightarrow{-\frac{1}{3}R2 \mapsto R2} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 7 & 9 & 3 \end{array} \right] \xrightarrow{R3-7R2 \mapsto R3} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & \frac{-1}{3} & \frac{16}{3} \end{array} \right] \\ &\xrightarrow{-3R3 \mapsto R3} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & \frac{4}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -16 \end{array} \right] \xrightarrow{R2-\frac{4}{3}R3 \mapsto R2} \left[ \begin{array}{ccc|c} 1 & -2 & -5 & 0 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \xrightarrow{R1+5R3 \mapsto R1} \left[ \begin{array}{ccc|c} 1 & -2 & 0 & -80 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \\ &\xrightarrow{R1+2R2 \mapsto R1} \left[ \begin{array}{ccc|c} 1 & 0 & 0 & -38 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & -16 \end{array} \right] \implies (x, y, z) = (-38, 21, -16) \end{aligned}$$

## 1.2 Gaussian Elimination

Gaussian elimination is a method for solving systems of linear equations by transforming the system's augmented matrix into a simpler form using elementary row operations. The goal is to reach **row echelon form** or **reduced row echelon form**, from which the solutions can be easily obtained.

### Row Echelon Form

**Definition** A matrix is in **row echelon form** if it satisfies the following conditions:

- All non-zero rows are above any rows of all zeros.
- The leading entry (the first non-zero number from the left, also called a **pivot**) of each non-zero row is to the right of the leading entry of the previous row.
- The leading entry in any non-zero row is 1 (this condition is sometimes omitted in basic row echelon form).

**Example** of a matrix in row echelon form:

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

It is often useful to further refine this to **reduced row echelon form**, which additionally requires that the leading 1 in each non-zero row is the only non-zero entry in its **column**.

**Example** of a matrix in reduced row echelon form:

$$\left[ \begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

### Notes

- Every matrix can be reduced to row echelon form and reduced row echelon form using a sequence of elementary row operations.
- The reduced row echelon form of any matrix is unique.
- Row echelon form is not unique; different sequences of row operations can lead to different row echelon forms for the same matrix.

**Zero Matrix** A **zero matrix** is a matrix in which all entries are zero. It is often denoted by 0. The solution to the zero matrix is  $\mathbb{R}^n$ , where n is the number of variables.

## Matrix Reduction Algorithm

1. Identify the leftmost non-zero column (the pivot column).
2. Select a non-zero entry in the pivot column as the pivot. If necessary, swap rows to move this entry to the top of the pivot column.
3. Scale the pivot row so that the pivot entry becomes 1.
4. Use row addition to create zeros in all positions below the pivot.
5. Cover the row containing the pivot and repeat the process for the submatrix that remains until all rows have been processed.
6. (Optional) To achieve reduced row echelon form, repeat the process to create zeros above each pivot as well.

## Gaussian Elimination Algorithm

1. Augmented Matrix  $[A|b]$  RREF with Row Operations
2. If a row of the form  $[0 \ 0 \ \dots \ 0 \ | \ b]$  with  $b$  non-zero exists, then the system is inconsistent (no solutions).
3. Otherwise, assign free variables (if any) and solve for leading variables to find the solution set.
4. Write leading variables in terms of the parameters
5. If there are no nonleading variables, we have 1 solution. If there are nonleading variables, we have infinitely many solutions.

**Example 1** Solve the system:

$$\begin{aligned}
 \begin{cases} 3x + y - 4z = 1 \\ x + 10z = 5 \\ 4x + y + 6z = 1 \end{cases} &= \left[ \begin{array}{ccc|c} 3 & 1 & -4 & 1 \\ 1 & 0 & 10 & 5 \\ 4 & 1 & 6 & 1 \end{array} \right] \xrightarrow{R2 \leftrightarrow R1} \left[ \begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & 1 \\ 4 & 1 & 6 & 1 \end{array} \right] \xrightarrow{R2-3R1 \rightarrow R2} \left[ \begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 4 & 1 & 6 & 1 \end{array} \right] \\
 &\xrightarrow{R3-4R1 \rightarrow R3} \left[ \begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 1 & -34 & -19 \end{array} \right] \xrightarrow{R3-R2 \rightarrow R3} \left[ \begin{array}{ccc|c} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -14 \\ 0 & 0 & 0 & -5 \end{array} \right] \implies \text{no solutions}
 \end{aligned}$$

**Example 2** Solve the system:

$$\begin{aligned}
 \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ 2x_1 - 4x_2 + x_3 = 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 = 4 \end{cases} &= \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array} \right] \xrightarrow{R2-2R1 \rightarrow R2} \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 1 & -2 & 2 & -3 & 4 \end{array} \right] \\
 &\xrightarrow{R3-R1 \rightarrow R3} \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array} \right] \xrightarrow{R3-R2 \rightarrow R3} \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \\
 &\xrightarrow{\frac{1}{3}R2 \rightarrow R2} \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] = \begin{cases} x_1 - 2x_2 - x_3 + 3x_4 = 1 \\ x_3 - 2x_4 = 1 \end{cases} \\
 &\Rightarrow \begin{cases} x_2 = s \\ x_4 = t \\ x_1 = 1 + 2s + t \\ x_3 = 1 + 2t \end{cases}
 \end{aligned}$$


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## Rank of a Matrix

**Definition** The **rank** of a matrix is the number of leading 1's (pivots) in its row echelon form or reduced row echelon form. It represents the dimension of the column space (or row space) of the matrix, which is the span of its columns (or rows). Rank is useful for finding the number of parameters in the solution set of a linear system.

**Example 3** Consider the matrix from Example 2. Find the rank and number of parameters:

$$\begin{aligned}
 A = \left[ \begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array} \right] &\xrightarrow{\text{RREF}(A)} \left[ \begin{array}{cccc|c} 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \text{rank}(A) = 2 \\
 n = 4 \text{ (number of variables)} &\Rightarrow \text{number of parameters} = n - \text{rank}(A) = 4 - 2 = 2
 \end{aligned}$$

**Example 4** Find the rank of the following matrices:

$$\begin{aligned}
 A = \left[ \begin{array}{cccccc} 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] &\Rightarrow \text{rank}(A) = 2 \\
 B = \left[ \begin{array}{cc} 0 & 0 \\ 0 & 3 \end{array} \right] &\xrightarrow{R1 \leftrightarrow R2} \left[ \begin{array}{cc} 0 & 3 \\ 0 & 0 \end{array} \right] \xrightarrow{\frac{1}{3}R1 \rightarrow R1} \left[ \begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \Rightarrow \text{rank}(B) = 1 \\
 C = \left[ \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] &\Rightarrow \text{rank}(C) = 0
 \end{aligned}$$



## 1.3 Homogeneous Systems

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**Definition** A system is called **homogeneous** if all the constant terms are zero. In other words, a system of equations is homogeneous if it can be written in the form

$$\begin{cases} x - 5y = 0 \\ 2x + 3y + z = 0 \end{cases}$$

The solution set of a homogeneous system always contains the **trivial solution**, which is the solution where all variables are equal to zero. For example, in the system above, the trivial solution is  $x = 0, y = 0, z = 0$ . Any other solution is called a **nontrivial solution**.

**Example** 1.3.1

**Theorem** If a homogeneous system of linear equations has more variables than equations, then it has infinitely many non-trivial solutions. For some matrix  $A$  with  $m$  rows and  $n$  parameters  $\text{rank}(A) \leq m < n$  implies that the solutions involved  $n - \text{rank}(A) \geq 1$  parameters, so  $x_i = S$  and we can choose any value for  $S$  other than zero to get a non-trivial solution.

**Note** If a homogeneous system has a nontrivial solution, it need not have more variables than equations.

**Note** If a homogeneous system has a non-trivial solution, then it has infinitely many non-trivial solutions. If  $x = S$  is a non-trivial solution, then so is  $x = kS$  for any scalar  $k \neq 0$ .

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## Vectors

**Definition** A **vector** in  $\mathbb{R}^n$  is an ordered  $n$ -tuple of real numbers. We can represent a vector as a column matrix:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = (x_1, x_2, \dots, x_n) = \langle x_1, x_2, \dots, x_n \rangle$$

where  $x_1, x_2, \dots, x_n$  are the components of the vector  $\vec{x}$ . In this class, vectors should always be represented as column matrices unless otherwise specified.

**Definition** A vector of length  $n$  is an ordered  $n$ -tuple, and the set of all vectors of length  $n$  is denoted by  $\mathbb{R}^n$ . In other words,  $\vec{x} \in \mathbb{R}^n$  if  $\vec{x}$  has  $n$  components.

**Vector Addition** Vector addition can only be performed on vectors of the same length. If  $\vec{x}, \vec{y} \in \mathbb{R}^n$ , then their sum is given by

$$\vec{x} + \vec{y} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

**Scalar Multiplication** If  $\vec{x} \in \mathbb{R}^n$  and  $c$  is a scalar, then the scalar multiplication of  $c$  and  $\vec{x}$  is given by

$$c\vec{x} = c \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$$

**Linear Combination** A sum of scalar multiples of several vectors or columns is called a **linear combination** of the vectors or columns. For example, if  $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$  and  $a, b, c$  are scalars, then

$$a\vec{u} + b\vec{v} + c\vec{w}$$

is a linear combination of the vectors  $\vec{u}, \vec{v}, \vec{w}$ .

### Example 1.3.2

**Note** To find if  $\vec{x}$  is a linear combination of  $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ , we can form the augmented matrix

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \mid \vec{x}]$$

and row reduce to see if there is a solution.